L3-Searching and Sorting

Algorithm	Best	Reason	Average	Reason	Worst	Reason	Space	Reason
Linear Search (Normal)	O(1)	First element-le kidaikkum.	O(n)	Half array check pannuvom.	O(n)	Ella elements check pannuvom.	O(1)	Extra variable mattum.
Linear Search (Recursion)	O(1)	First call-le kidaikkum.	O(n)	Half recursive calls.	O(n)	Ella recursive calls pogum.	O(n)	Recursion stack use aagum.
Binary Search (Normal)	O(1)	Middle-le kidaikkum.	O(log n)	Half-half search pannuvom.	O(log n)	Half-half reduce pannuvom.	O(1)	Extra variable mattum.
Binary Search (Recursion)	O(1)	First middle-le kidaikkum.	O(log n)	Half-half recursion.	O(log n)	Last recursion vara pogum.	O(log n)	Recursion stack.
Selection Sort	$O(n^2)$	Ella pass-um compare pannuvom.	$O(n^2)$	Ella pass-um compare pannuvom.	$O(n^2)$	Ella pass-um compare pannuvom.	O(1)	Extra array illa.
Bubble Sort	O(n)	Already sorted.	$O(n^2)$	Neraya swaps.	$O(n^2)$	Maximum swaps.	O(1)	Extra array illa.
Insertion Sort	O(n)	Already sorted.	$O(n^2)$	Elements shift aagum.	$O(n^2)$	Maximum shifting.	O(1)	One temp variable.
Shell Sort	O(n log n)	Gap use pannuvom.	O(n^1.5)	Gap depend pannum.	$O(n^2)$	Bad gap sequence.	O(1)	Extra array illa.
Merge Sort	O(n log n)	Divide & Merge.	O(n log n)	Divide & Merge.	O(n log n)	Divide & Merge.	O(n)	Temporary array use pannuvom.
Quick Sort	O(n log n)	Good pivot.	O(n log n)	Balanced partition.	$O(n^2)$	Bad pivot.	O(log n) Avg / O(n) Worst	Recursion stack.
Counting Sort	O(n+k)	Counting pannuvom.	O(n+k)	Counting pannuvom.	O(n+k)	Counting pannuvom.	O(n+k)	Count array use pannuvom.
Radix Sort	O(d(n+k))	Digit by digit sort.	O(d(n+k))	Digit by digit sort.	O(d(n+k))	Digit by digit sort.	O(n+k)	Count array use pannuvom.
Heap Sort	O(n log n)	Heap use pannuvom.	O(n log n)	Heap use pannuvom.	O(n log n)	Heap use pannuvom.	O(1)	Extra array illa.
Bucket Sort	O(n+k)	Equal buckets.	O(n+k)	Equal buckets.	$O(n^2)$	Oru bucket-la ellam pogum.	O(n+k)	Bucket array use pannuvom.